

# Bases and Integral Relations for the $k$ -Kernel of Euler's Totient

*Exact finite-level ranks, with supplementary results on Erdős Problem #249*

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## ABSTRACT

For integers  $k \geq 2$  and  $e \geq 1$ , we give a basis for the rational span of the totient sections  $n \mapsto \varphi(k^j n + r)$  with  $0 \leq j \leq e$  and  $0 \leq r < k^j$ . Its dimension is  $k^e + 1$ . Every section is an explicit integer multiple of a retained one, and these reductions generate all integral relations, with unique coefficients. The proof combines the local formula for Euler's totient with a nonsingular evaluation matrix obtained from the Chinese remainder theorem and Dirichlet's theorem. Supplementary results concern bounded residue series and conditions for irrationality of  $S = \sum_{n \geq 1} \varphi(n)2^{-n}$ . They do not establish that  $S$  is irrational.

*Formal proofs.* A result with a kernel-checked Lean proof carries a mark in the margin. *Lean* opens the proof: the declaration itself when one declaration states the whole result, otherwise the list of declarations that together state it. *Comparator* opens the record of an independent check, in which the same statement, written again from Mathlib alone in a separate repository, was compared with our proof by Lean's Comparator tool, allowing only the three standard axioms. A dagger on the Lean mark means that the Lean proof assumes an input named just below the result. A result without a mark has no Lean proof of its whole statement; what is checked is said below it. The [evidence record](#) gives every declaration, version and check. These checks show that the stated propositions are proved; whether they are the right propositions is a question the reader can settle by comparing them with the text. Equivalent formulation [3.2](#) carries no mark and is proved in Lean; the link follows its proof.

## 1 A basis and all its relations

We write  $\varphi$  for Euler's totient function and use  $\mathbb{N} = \{0, 1, 2, \dots\}$ , with  $\varphi(0) = 0$ . For  $k \geq 2$ , the  $k$ -kernel consists of the sequences  $n \mapsto \varphi(k^j n + r)$  with  $j \geq 0$  and  $0 \leq r < k^j$ . Some of these sections are scalar multiples of others: for example,  $\varphi(4n + 2) = \varphi(2n + 1)$  and  $\varphi(4n) = 2\varphi(2n)$ . Thus the seven sections through level two in base two are spanned by

$$n \mapsto \varphi(n), \quad \varphi(2n), \quad \varphi(2n + 1), \quad \varphi(4n + 1), \quad \varphi(4n + 3).$$

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\**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

Are these five sequences independent, and do the analogous reductions account for every relation in every base? The next theorem answers both questions; Corollary 1.3 gives the integral relation module.

An integer sequence is  $k$ -regular if the  $\mathbb{Z}$ -module generated by its  $k$ -kernel is finitely generated [1]. Coons proved that the totient sequence is not  $k$ -regular for any  $k \geq 2$  [2, Thm. 3.2]. Martin's Theorem 1 already implies the affine independence used below [4, Thm. 1]. The theorem below identifies the finite-level basis and all integral relations.

**Theorem 1.1** (A basis through each finite level). *Let  $k \geq 2$  and  $e \geq 1$  be integers, write  $F_{j,r}^{(k)}(n) = \varphi(k^j n + r)$ , and put*

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$$V_{k,e} = \text{span}_{\mathbb{Q}} \{ F_{j,r}^{(k)} : 0 \leq j \leq e, 0 \leq r < k^j \}.$$

Then  $\dim_{\mathbb{Q}} V_{k,e} = k^e + 1$ , and

$$\mathcal{B}_{k,e} = \{F_{0,0}^{(k)}, F_{1,0}^{(k)}\} \cup \{F_{j,r}^{(k)} : 1 \leq j \leq e, 1 \leq r < k^j, k \nmid r\}$$

is a basis. Every omitted section reduces to a basis element by an explicit scalar:  $F_{j,0}^{(k)} = k^{j-1} F_{1,0}^{(k)}$  for  $j \geq 1$ , and if  $r = k^t u$  with  $t = \max\{s : k^s \mid r\} \geq 1$  and  $k \nmid u$ , then

$$F_{j,r}^{(k)} = C_k(t, u) F_{j-t, u}^{(k)}, \quad C_k(t, u) = k^t \prod_{\substack{p \mid k \\ p \nmid u}} \left(1 - \frac{1}{p}\right).$$

The condition on a retained positive residue is  $k \nmid r$ , not  $\gcd(k, r) = 1$ . Thus  $r = 2$  is retained when  $k = 6$ . No primality or squarefreeness assumption on the base is needed. The reduction scalar can depend on the residue even for a fixed base: at  $k = 6$ ,

$$\varphi(36n + 6) = 2\varphi(6n + 1), \quad \varphi(36n + 12) = 4\varphi(6n + 2).$$

The second scalar differs because the argument  $6n + 2$  already contains the prime 2.

## 1.1 Proof of the basis theorem

First consider spanning. The zero-residue reduction follows from  $\varphi(k^j n) = k^{j-1} \varphi(kn)$  for  $j \geq 1$ : the prime divisors are unchanged when  $n > 0$ , and both sides vanish at  $n = 0$ . For a positive residue and a prime  $p \mid k$  with  $p \nmid u$ , the argument  $k^{j-t} n + u$  is a  $p$ -adic unit. Multiplication by  $k^t$  therefore contributes its power of  $p$  and one factor  $1 - 1/p$  to the totient. When  $p \mid u$ , the argument already contains  $p$ , and only the power contributes. This gives  $C_k(t, u)$  and reduces every omitted section to the displayed family.

For independence, we make an evaluation matrix diagonal and nonzero modulo a prime. Each row requires just one primitive affine form to take a prime value; auxiliary prime divisors make the other totients vanish modulo the chosen prime. The two zero-residue sections are treated separately at the end.

Consider  $s$  forms  $L_i(n) = a_i n + b_i$ ,  $1 \leq i \leq s$ , with integer coefficients, positive slopes and  $a_i b_j \neq a_j b_i$  for  $i \neq j$ . We may restrict to a tail and shift the variable so that every intercept is positive. Set  $g_i = \gcd(a_i, b_i)$  and  $A_i = L_i/g_i$ ; the slope and intercept of  $A_i$  are coprime. Choose an odd prime  $\ell$  avoiding the slopes, all  $g_i \varphi(g_i)$ , and the nonzero

cross determinants. In row  $i$ , choose distinct primes  $q_{ij} \equiv 1 \pmod{\ell}$  for  $j \neq i$ , outside the same finite exceptional set, and impose

$$L_j(n) \equiv 0 \pmod{q_{ij}}, \quad A_i(n) \equiv 2 \pmod{\ell}.$$

The Chinese remainder theorem combines these congruences. The progression for  $A_i(n)$  has modulus  $(a_i/g_i)\ell \prod_{j \neq i} q_{ij}$  and is reduced. Primitivity gives  $\gcd(A_i(n), a_i/g_i) = 1$  for every integer  $n$ , and the residue 2 excludes  $\ell$ . If  $q_{ij}$  divided  $A_i(n)$ , then it would divide both  $L_i(n)$  and  $L_j(n)$ , hence their nonzero cross determinant, contrary to its choice. Dirichlet's theorem supplies arbitrarily large primes  $p = A_i(n)$  with  $p \nmid g_i$ . Since  $L_i(n) = g_i p$ , the diagonal value is  $\varphi(L_i(n)) = \varphi(g_i)(p-1) \not\equiv 0 \pmod{\ell}$ . Each off-diagonal value is divisible by  $\ell$ , since  $q_{ij} - 1$  divides the corresponding totient. Choose such an evaluation point  $n_i$  for each row  $i$ . The matrix  $(\varphi(L_j(n_i)))_{i,j}$  is diagonal modulo  $\ell$ , with every diagonal entry nonzero. Its determinant is nonzero modulo  $\ell$ , hence is a nonzero integer. This proves rational linear independence.

To handle the two proportional zero-residue forms, restrict to  $n = km + 1$ , where  $\varphi(kn) = \varphi(k)\varphi(n)$  by coprimality. The positive-residue sections then give pairwise non-proportional forms

$$k^{j+1}m + (k^j + r), \quad k \nmid r,$$

and the remaining form is  $km + 1$ . Proportionality between two of the positive-residue forms would give  $r/k^j = r'/k^{j'}$ . If  $j' > j$ , then  $r' = k^{j'-j}r$  is divisible by  $k$ , contrary to the choice of  $r'$ ; if  $j = j'$ , then  $r = r'$ . None is proportional to  $km + 1$ , since  $r > 0$ . The matrix therefore eliminates every positive-residue coefficient. If  $A, B$  are the two zero-residue coefficients, the restriction gives  $A + B\varphi(k) = 0$ . Evaluation at  $n = k$  gives  $A + Bk = 0$ , since  $\varphi(k^2) = k\varphi(k)$ . As  $\varphi(k) < k$ , both coefficients vanish. The reductions give spanning and the count is

$$2 + \sum_{j=1}^e (k^j - k^{j-1}) = k^e + 1.$$

This proves Theorem 1.1.

The relation to the earlier CRT argument and the alternative proof from Martin's theorem are discussed in Appendix A.

**Corollary 1.2** (Dyadic basis). *The family*

$$\{\varphi_{0,0}, \varphi_{1,0}\} \cup \{\varphi_{j,r} : j \geq 1, 0 < r < 2^j, r \text{ odd}\}, \quad \varphi_{j,r}(n) = \varphi(2^j n + r),$$

*is a basis for the rational span of the full dyadic kernel. Every rational relation is generated by the scalar reductions. The complete level-zero truncation has dimension one.*

*Proof.* Every finite subset of the displayed family lies in  $\mathcal{B}_{2,e}$  for some  $e \geq 1$ , so the theorem gives independence. The reductions give spanning. At level zero the only section is  $\varphi(n)$ , which is nonzero since  $\varphi(1) = 1$ , so its span has dimension one.  $\square$

**Corollary 1.3** (Integral coordinates and all integral relations). *Let  $k \geq 2$  and  $e \geq 1$ . The retained family is a  $\mathbb{Z}$ -basis of the module generated by the sections through level  $e$ . Index*

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the sections by their level and residue, retaining distinct indices even when they define equal sequences. For each omitted index  $i$ , write the scalar reduction as  $F_i = a_i F_{j(i)}$ , where  $j(i)$  is retained and  $a_i$  is a nonnegative integer. In the free abelian group with one generator  $E_i$  for each of these indices, the vectors

$$R_i = E_i - a_i E_{j(i)} \quad (i \text{ omitted})$$

form a  $\mathbb{Z}$ -basis of the kernel of evaluation  $E_i \mapsto F_i$ . In particular, every integral relation has a unique integral expression in these elementary relations, and their rank is  $\sum_{j=1}^{e-1} k^j$ .

*Proof.* The denominator  $\prod_{p|k, p \nmid u} p$  in  $C_k(t, u)$  divides  $k$ , so the successive section reductions have integer scalars. They give integral coordinates in the retained family; its rational independence makes these coordinates unique over  $\mathbb{Z}$  as well.

Now let  $x = \sum_i x_i E_i$  evaluate to zero. For each omitted index  $i$ , subtract  $x_i R_i$ . The result is supported on retained indices and still evaluates to zero, so independence forces it to vanish. Thus the  $R_i$  generate the integral relation module. Conversely, the coefficient at an omitted index  $i$  in  $\sum_h c_h R_h$  is exactly  $c_i$ : each relation has coefficient one at its own omitted coordinate and zero at all the others. This proves independence and uniqueness without dividing by any integer. There are

$$(1 + k + \cdots + k^e) - (k^e + 1) = \sum_{j=1}^{e-1} k^j$$

omitted sections. □

For example, at base 2 and depth 2, retain  $F_{0,0}, F_{1,0}, F_{1,1}, F_{2,1}, F_{2,3}$ . The two omitted sections satisfy  $F_{2,0} = 2F_{1,0}$  and  $F_{2,2} = F_{1,1}$ , so

$$E_{2,0} - 2E_{1,0}, \quad E_{2,2} - E_{1,1}$$

are an integral basis of all relations among the seven sections. The rank calculation alone would give two; the basis also identifies every relation and its unique coefficients.

The assertion concerns the integer combinations of the original sections, not every integer-valued sequence in their rational span. For example, at base 2 and  $e \geq 2$ ,  $\varphi(4n+3)/2$  takes integer values but has coefficient  $1/2$  in the retained basis. The generated module is therefore not saturated in the group of integer-valued sequences: an integer-valued sequence outside the module can have a nonzero integer multiple inside it.

## 1.2 Relations with periodic coefficients

Pairwise nonproportionality is essential here. For instance,  $n+1$  and  $n+2$  satisfy the hypothesis. In contrast,  $4n+2$  and  $2n+1$  are proportional, and  $\varphi(4n+2) = \varphi(2n+1)$  gives a nontrivial relation. Periodicity imposes no bound on the common period: it permits any finite periodic pattern, because one can work on each residue class separately.

**Corollary 1.4** (Periodic coefficients). *Let  $L_1, \dots, L_s$  be pairwise nonproportional affine forms with integer coefficients and positive slopes. If  $w_1, \dots, w_s$  are rational-valued periodic sequences, then*

$$\sum_i w_i(n) \varphi(L_i(n)) = 0 \quad \text{for all sufficiently large } n \implies w_i(n) = 0 \quad \text{for every } i, n.$$

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*Proof.* Write  $L_i(n) = a_i n + b_i$  and choose a common positive period  $Q$  for the  $w_i$ . On a progression  $n = r + Qt$ , with  $0 \leq r < Q$ , their values are the constants  $w_i(r)$ . The restricted affine forms have cross determinants  $Q(a_i b_j - a_j b_i) \neq 0$ . Shifting  $t$  beyond the stated threshold preserves these determinants and makes all arguments positive. Affine independence then gives  $w_i(r) = 0$  for every  $i$ . Doing this for each residue  $r$  proves the conclusion at all indices, by periodicity.  $\square$

Applied on each progression, Martin's Theorem 1 gives the same conclusion [4, Thm. 1]. The same argument preserves the basis after deletion of a finite prefix. For the two sections with residue zero, large  $n \equiv 1 \pmod{k}$  and large powers of  $k$  replace the two evaluations used above.

**Scope.** This completes the basis argument. It describes relations among coefficient subsequences, not the arithmetic nature of  $S = \sum_{n \geq 1} \varphi(n) 2^{-n}$ .

## Supplementary results on the binary totient series

The independent results below distinguish bounded-residue series, finite exclusions for  $S$ , and conditions for irrationality. None is needed for Theorem 1.1; in particular, the conditions in Sections 3 and 5 are not assumptions of that theorem. Generating-function context and the formal-source guide are in Appendix A.

## 2 Bounded residues and rationality

After reducing  $\varphi(n)$  modulo a fixed integer, the coefficients are bounded. A single nonzero coefficient between long zero blocks then contradicts rationality by an elementary tail estimate.

**Theorem 2.1** (Residue series and dyadic observables). *For every  $m \geq 3$ ,*

$$\sum_{n \geq 1} \frac{\varphi(n) \bmod m}{2^n} \notin \mathbb{Q}.$$

*For  $k \geq 1$  and  $f : \mathbb{Z}/2^k\mathbb{Z} \rightarrow \mathbb{Q}$ , the series  $\sum_{n \geq 1} f(\varphi(n) \bmod 2^k) 2^{-n}$  is rational exactly when  $f$  is constant on the even residue classes. If that constant is  $c$ , its value is  $3f(1)/4 + c/4$ .*

The restriction  $m \geq 3$  is necessary: modulo 2 the only nonzero coefficients occur at  $n = 1, 2$ , and the sum is  $1/2 + 1/4 = 3/4$ . For the second assertion, only  $f(1)$  and its values on the even classes can affect the sum. The other odd classes are never attained by  $\varphi(n)$ .

The next lemma applies, for example, to the indicator of the squares: its coefficients are bounded, and sufficiently large squares have no other square within any prescribed distance. An eventually periodic sequence with infinitely many nonzero terms cannot satisfy the isolated-coefficient hypothesis. Most importantly, the bound  $C$  must be fixed before the zero blocks are chosen; the unbounded coefficients  $\varphi(n)$  do not satisfy this hypothesis.

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**Lemma 2.2** (An isolated nonzero coefficient between long zero blocks). *Let  $a_n \in \mathbb{Z}$  satisfy  $|a_n| \leq C$ . Suppose that for arbitrarily large  $L$  there is  $N > L$  such that  $a_N \neq 0$  and  $a_{N+t} = 0$  for  $0 < |t| \leq L$ . Then  $\sum_{n \geq 1} a_n 2^{-n}$  is irrational.*

*Proof.* If the sum has denominator  $q$ , every scaled tail  $T_j = \sum_{t \geq 1} a_{j+t} 2^{-t}$  belongs to  $q^{-1}\mathbb{Z}$ . Choose  $L$  so large that  $C2^{-L} < 1/q$ . The right zero block gives  $|T_N| \leq C2^{-L} < 1/q$ , hence  $T_N = 0$ . The left zero block now gives  $T_{N-L-1} = a_N 2^{-L-1}$ , a nonzero element of  $q^{-1}\mathbb{Z}$  with absolute value less than  $1/q$ , a contradiction.  $\square$

*Proof of Theorem 2.1.* Fix a unit  $s \pmod{m}$  and a block length  $L$ . For each nonzero  $t \in [-L, L]$ , choose a distinct prime  $q_t \equiv 1 \pmod{m}$  larger than  $L$  and outside the prime divisors of  $m$ . Prescribe

$$p \equiv s \pmod{m}, \quad p \equiv -t \pmod{q_t}.$$

This is a reduced progression, because each  $q_t > |t|$ . Dirichlet supplies arbitrarily large primes  $p$  in it. Then  $m \mid \varphi(p+t)$  for every  $0 < |t| \leq L$ , whereas  $\varphi(p) \equiv s-1 \pmod{m}$ .

Now let  $f: \mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{Q}$ . Subtract  $f(0)$  and clear the finitely many denominators of its values. This produces a bounded integer sequence. If  $f(s-1) \neq f(0)$ , the construction gives an isolated nonzero coefficient between arbitrarily long zero blocks, so Lemma 2.2 proves irrationality. For the least-residue map choose  $s = -1$ : its central residue is  $m-2 \neq 0$  when  $m \geq 3$ . For  $m = 2^k$ , every even residue  $r$  has  $r+1$  a unit, so a nonconstant restriction to the even residues supplies such a nonzero coefficient. Conversely,  $\varphi(n)$  is even for every  $n \geq 3$ . Constancy on the even classes makes all terms after  $n = 2$  constant, giving the stated rational value without a recurrence hypothesis.  $\square$

There is an earlier coefficient-level obstruction: for each  $m \geq 3$ ,  $\varphi(n) \pmod{m}$  is not automatic in any integer base  $k \geq 2$  [10, Cor. 4, p. 654]; see also [11, Thm. 3 and Ex. 4]. Nonautomaticity does not prove irrationality of this sum: when  $m > 2$ , the residues are not binary digits. The preceding proof instead uses the isolated coefficient and the bounded tail. Erick Wong posted a proof that  $\sum_{n \geq 1} (\varphi(n) \pmod{k}) k^{-n}$  is irrational for every integer  $k > 2$ , with the base equal to the modulus, as an answer on Mathematics Stack Exchange on 29 March 2015 [5]. Theorem 2.1 fixes the base at 2 for every modulus  $m \geq 3$  and classifies the dyadic residue observables.

The preceding proof fixes its coefficient bound before choosing the zero blocks. For  $S$ , the termwise condition  $2^L \mid \varphi(N+L)$ , with  $N+L > 1$ , forces  $2^L \leq \varphi(N+L) \leq N+L-1$ . Hence the tail envelope  $(N+L+2)2^{-L} > 1$  cannot exclude any positive denominator. The following criterion instead measures cancellation in a weighted prefix.

### 3 Tail differences and finite residue tests

Write

$$S = \sum_{n \geq 1} \varphi(n) 2^{-n}, \quad R_N = \sum_{j \geq 1} \varphi(N+j) 2^{-j}, \quad \Delta_h(N) = R_{N+h} - R_N.$$

The identity

$$2^N S = \sum_{n=1}^N \varphi(n) 2^{N-n} + R_N$$

has an integer finite sum. Subtracting the identities at  $N + h$  and  $N$  gives

$$\Delta_h(N) \equiv 2^N(2^h - 1)S \pmod{\mathbb{Z}}. \quad (1)$$

In particular, a nonintegral tail difference excludes every rational value whose denominator divides  $2^N(2^h - 1)$ .

### 3.1 Certificates at arbitrarily large multiples of a shift

For  $h, N, L \in \mathbb{N}$ , let

$$D_{h,N,L} = \sum_{j=0}^{L-1} (\varphi(N + h + 1 + j) - \varphi(N + 1 + j))2^{L-1-j}, \quad B = N + h + L + 2.$$

All residues below are least nonnegative residues. Write  $K(h, N, L)$  for

$$B < D_{h,N,L} \pmod{2^L} < 2^L - B.$$

The omitted terms form another difference of scaled tails:

$$2^L \Delta_h(N) - D_{h,N,L} = R_{N+h+L} - R_{N+L}.$$

Since  $0 \leq R_M \leq \sum_{j \geq 1} (M + j)2^{-j} = M + 2$ , the absolute difference is bounded by the larger of the two tail bounds, not their sum. Thus

$$|2^L \Delta_h(N) - D_{h,N,L}| \leq B. \quad (2)$$

A certificate  $K(h, N, L)$  therefore forces  $\Delta_h(N) \notin \mathbb{Z}$ . Conversely, write  $\|x\|_{\mathbb{R}/\mathbb{Z}} = \min_{z \in \mathbb{Z}} |x - z|$  for the distance to the nearest integer. If  $\Delta_h(N) \notin \mathbb{Z}$ , choose  $L$  so that  $2^L \|\Delta_h(N)\|_{\mathbb{R}/\mathbb{Z}} > 2B$ . Such a choice exists:  $h, N$  are fixed, the distance is positive, and  $B = N + h + L + 2$  grows only linearly in  $L$ . The error estimate then puts the residue of  $D_{h,N,L}$  more than  $B$  from either endpoint. Thus nonintegrality is equivalent to a certificate at some unrestricted depth.

**Theorem 3.1** (Propagation of one nonintegral tail difference). *Fix  $d \geq 1$  and  $N \geq 0$ . If  $\Delta_d(N) \notin \mathbb{Z}$ , then every sufficiently late pair  $\{t, t + 1\}$  contains an  $m$  such that  $K(md, N, md)$  holds. Consequently*

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$$\exists t \geq 1 : K(td, N, td) \iff \Delta_d(N) \notin \mathbb{Z}.$$

Requiring this for every  $d \geq 1, N \geq 0$  is equivalent to  $S \notin \mathbb{Q}$ .

For a fixed  $d, N$ , the hypothesis is just one nonintegral tail difference; it does not assert irrationality of  $S$ . Requiring it for every  $d, N$  is as strong as the original problem. The conclusion sets the truncation depth equal to the shift,  $L = md$ , and loses at most one member of each sufficiently late pair of successive multipliers.

*Proof.* Set  $F_t = \Delta_{td}(N)$ . Equation (1) yields  $F_{t+1} \equiv 2^d F_t + F_1 \pmod{\mathbb{Z}}$ . If  $K(td, N, td)$  fails, (2) gives

$$\|F_t\|_{\mathbb{R}/\mathbb{Z}} \leq \epsilon_t, \quad \epsilon_t = 2(N + 2td + 2)2^{-td}.$$

Two adjacent failures would imply  $\|F_1\|_{\mathbb{R}/\mathbb{Z}} \leq 2^d \epsilon_t + \epsilon_{t+1}$ , impossible for all sufficiently large  $t$  because the left side is positive and the right side tends to zero. Conversely, if  $F_1$  is integral, then every  $F_t$  is integral by the same recurrence, so no certificate exists. Finally, (1) is nonintegral for every  $d, N$  when  $S$  is irrational. If  $S = a/(2^c v)$  with  $v$  odd, take  $N = c$  and  $d = \varphi(v)$  to make it integral.  $\square$

For instance  $D_{1,12,16} = -143140$ , with residue 53468 modulo  $2^{16}$  and  $B = 31$ , verifies the hypothesis at  $d = 1, N = 12$ . The theorem gives infinitely many certificates with depth equal to the shift for this pair, not for every  $d, N$ .

### 3.2 An equivalent residue condition for each hypothetical denominator

For  $c \geq 0$ , odd  $v \geq 1$  and a positive multiple  $H$  of  $\varphi(v)$ , put

$$M = \frac{2^H - 1}{v}, \quad B_{H,c} = \sum_{j=0}^{H-1} \varphi(c+1+j) 2^{H-1-j}.$$

Euler's theorem makes  $M$  integral.

*Equivalent formulation 3.2* (A residue test modulo a divisor of  $2^H - 1$ ). The series  $S$  is irrational if and only if, for every  $c \geq 0$  and odd  $v \geq 1$ , some positive multiple  $H$  of  $\varphi(v)$  satisfies

$$c + H + 1 < (-B_{H,c}) \bmod M < M - (c + H + 1). \quad (3)$$

*Proof.* The identity

$$B_{H,c} = 2^H R_c - R_{c+H} = M(vR_c) - \Delta_H(c)$$

is exact. For  $H \geq 1$ ,  $|\Delta_H(c)| < c + H + 1$ . Indeed  $R_N \leq N + 1$  for  $N \geq 1$  and  $0 < R_0 \leq 3/2$ ; the strict inequalities follow by subtracting a positive tail. If  $S = a/(2^c v)$ , then  $vR_c \in \mathbb{Z}$  and  $\Delta_H(c) \in \mathbb{Z}$ . The residue of  $-B_{H,c}$  is therefore within distance less than  $c + H + 1$  of 0 (mod  $M$ ), so (3) fails.

If  $S$  is irrational, fix  $c, v$ . Then  $vR_c$  is nonintegral. As  $H$  tends to infinity through positive multiples of  $\varphi(v)$ ,  $\Delta_H(c)/M \rightarrow 0$  and  $(c + H + 1)/M \rightarrow 0$ . Hence the fractional part of  $-B_{H,c}/M = -vR_c + \Delta_H(c)/M$  stays away from both endpoints of  $[0, 1]$ , which gives (3) for all sufficiently large such  $H$ .  $\square$

The condition is neither a routine size bound nor an additional assumption known for a broad class containing  $S$ : with its universal quantifiers it is equivalent to the unresolved irrationality assertion. A rational value  $a/(2^c v)$  fails the test for that same  $c, v$ , whereas irrationality gives it for every sufficiently large allowed  $H$ . The proof does not find such an  $H$  without assuming irrationality.

This equivalence is [proved in Lean](#), with the modulus written as  $vM = 2^H - 1$ . The finite residues also satisfy

$$\rho_{H,N+1,M} = (2\rho_{H,N,M} - \varphi(N+H+1) + \varphi(N+1)) \bmod M, \quad \rho_{H,N,M} = (-B_{H,N}) \bmod M,$$

which is an exact recurrence for searches directed at (3).

## 4 Information that does not force the gap

1. **Tail differences at least common multiples.** Let  $H_a = \text{lcm}(1, \dots, 2^a)$ . Then  $S$  is irrational if and only if, beyond every threshold  $a_0$ , some  $a \geq a_0$  satisfies  $\Delta_{H_a}(H_a) \notin \mathbb{Z}$ . This is the [exact cofinal nonintegrality criterion](#) for the actual LCM diagonal, not a quantitative separation estimate and not a proof that such indices occur; the [forward implication](#) is recorded separately.
2. **Positive tail differences need not be nonintegral.** For  $a \geq 8$  and  $J + (a + 6) < 2 \cdot 2^a$ , one has  $\Delta_{H_a}(H_a + J) > 0$ . Suppose this difference is an integer. If a truncation depth  $K$  satisfies

$$J + K + (a + 6) < 2 \cdot 2^a, \quad 2H_a + J + K + 2 < 2^K,$$

then

$$D_{H_a, H_a + J, K} \bmod 2^K = 2^K - \Delta_{H_a}(H_a + J + K)$$

lies strictly between  $2^K - (2H_a + J + K + 2)$  and  $2^K$ . The later tail difference is a positive integer; its *negative* is the representative near zero. Thus positivity determines which endpoint contains the residue, but does not exclude an integral tail difference. Both displayed bounds on  $K$  are required for this conclusion. *Checked: [positive tail differences](#), [negative endpoint representative](#), [residue near the upper endpoint](#).*

3. **Divisibility and finite linear combinations do not suffice.** A dyadic coboundary is a sequence of the form  $2u_j - u_{j+1}$ ; its weighted sum telescopes. For every  $t \geq 3$ , with  $H = \text{lcm}(1, \dots, t)$ , the linked source constructs a nonzero comparison sequence of this form. Its coefficients reproduce the actual totient differences at  $t-2$  prescribed indices and are divisible by  $\varphi(H)$ . Every finite integer linear combination of forward shifts preserves the telescoping identity and divisibility by  $\varphi(H)$  and the corresponding lower-level factors. The bounds remain uniform in the index, but are multiplied by the sum of the absolute values of the combination's coefficients. A variant preserves sparse prescribed values along entire arithmetic progressions. These properties alone do not force a contradiction: the comparison need not consist of actual totient differences. The theorem concerns linear combinations of shifts, not nonlinear operations, and does not supply certificates at arbitrarily large scales. *Checked: [obstruction for finite integer combinations of shifts](#).*

If  $S = a/v$  with integers  $a$  and  $v \geq 1$ , the scaled tails  $u_N = vR_N$  are integers and satisfy

$$u_{N+1} = 2u_N - v\varphi(N+1), \quad 0 \leq u_N \leq v(N+2).$$

In particular  $u_N/2^N \rightarrow 0$ , the condition called temperedness in the formal sources. Its dyadic sections  $n \mapsto u_{2^j n + r}$  have rational rank at least  $2^e - 1$  through level  $e$  and a common eventual period modulo  $v$ . Both properties are [proved in Lean for the same carry](#); they are compatible, and neither supplies a contradiction.

The companion, *The Binary Totient Series*, constructs a rational comparison sequence with  $c(n) = \varphi(n)$  for odd  $n$ ,  $|c(n) - \varphi(n)| \leq 2$  for even  $n$ ,  $0 \leq c(n) \leq n$ , and

$$\sum_{n \geq 1} c(n)2^{-n} = \frac{5}{4}.$$

This comparison sequence has an integral tempered carry with dyadic rank at least  $2^e - 1$  at every level. Thus the same lower bound for a hypothetical totient carry is compatible with rationality. For one fixed prime  $p$  and  $g : \mathbb{N} \rightarrow \mathbb{Z}$ , the exact laws  $g(pn) = pg(n)$  when  $p \mid n$  and  $g(pn) = (p-1)g(n)$  otherwise, together with  $g - \varphi = o(n)$ , force  $g = \varphi$ : a nonzero defect keeps a nonzero ratio to the argument along  $p^j n$ . The comparison sequence fails these dilation laws. The unchanged odd-residue sections also give coefficient rank at least  $2^e - 1$  at every level. This is a lower bound, not an exact formula for the full coefficient rank.

A different obstruction concerns rational approximation. Let  $\mu$  be the Möbius function and put

$$\Theta_2 = \sum_{d \geq 1} \frac{\mu(d)}{(2^d - 1)^2} = S - \frac{1}{2}, \quad Q(e, Y) = \frac{(\sum_{d=1}^Y \mu(d)/(2^d - 1)^{e+2})^2}{\sum_{d=1}^Y \mu(d)/(2^d - 1)^{2e+2}}.$$

The divisor-convolution identity  $\varphi = \mu * \text{Id}$  gives  $S = \sum_{d \geq 1} \mu(d)2^d/(2^d - 1)^2$ . Now  $2^d = (2^d - 1) + 1$  and  $\sum_{d \geq 1} \mu(d)/(2^d - 1) = 1/2$ , the latter following from  $\sum_{d \mid n} \mu(d) = 0$  for  $n > 1$  and 1 for  $n = 1$ . These identities give the displayed value of  $\Theta_2$ ; all the sums are absolutely convergent. Amiram Eldar posted the corresponding formula  $S = \frac{1}{2} + \sum_{d \geq 1} \mu(d)/(2^d - 1)^2$ , together with its coprimality interpretation, to OEIS A256936 on 15 March 2026 [6], and Steve Fan posted the formula on the problem's forum thread on 16 May 2026 [7]. The companion proves the lower bound  $1/480$  for these quotients; the following theorem gives the exact extremum.

**Theorem 4.1** (A lower bound for the rank-one quotients). *For  $e \geq 1$  and  $Y \geq 4$ , the denominator of  $Q(e, Y)$  is positive, and the unique minimiser is  $(e, Y) = (1, 5)$ . Every admissible quotient and every nonempty finite positive weighted average of such quotients satisfies*

[Lean](#)  
[Comparator](#)

$$Q - \Theta_2 > \frac{21}{320}.$$

The uniform bound  $Q - \Theta_2 > 1/15$  is false already at  $(1, 5)$ .

This theorem treats averages after the quotients are formed. Signed coefficients, coupling before the quotient, and higher-rank kernels require separate arguments. For rational approximants  $p_j/q_j$  in lowest terms, with  $q_j > 0$ , a sufficient arithmetic target is

$$0 < q_j |\Theta_2 - p_j/q_j| \rightarrow 0.$$

Convergence alone does not give this estimate. The elementary dyadic prefix bounds give only  $q_j |S - p_j/q_j| \leq N + 2$  at depth  $N$ , not a quantity tending to zero. A prescribed approximation family therefore needs a sharper error or reduced-denominator bound. When approximants may be chosen freely, continued-fraction convergents show that the criterion is equivalent to irrationality, just as (3) is. The theorem above excludes the displayed rank-one quotients and their positive averages, not other approximation families.

## 5 A remaining exponential-sum estimate

A sufficient analytic target is a bound on an explicit finite sum: for every  $h \geq 1$  and every  $X_0$ , find integers  $X \geq \max(X_0, 1)$  and  $L \geq h$  such that

$$16(2X + h + L + 2) \leq 2^L, \quad \sum_{N=X}^{2X-1} \cos\left(\frac{2\pi D_{h,N,L}}{2^L}\right) \leq \frac{9}{10}X.$$

Here  $D_{h,N,L}$  is the integer window sum from Section 3. The room inequality makes the truncation error small, whereas the cosine bound forces at least one residue away from the integer lattice. The companion proves that these quantified inequalities would imply irrationality. The modulus grows with  $L$ , and the coefficients inside the phase are totient differences; ordinary information about the distribution of prime factors does not give this estimate.

One proposed approach separates a single totient value from each phase. The following definitions specify the four estimates it would require. Fix integers  $h, s \geq 1$ ,  $L \geq s + h$ ,  $X \geq 1$  and  $0 < \eta < 1$ . Put  $t = L - s + 1$  and  $E_N = \exp(2\pi i D_{h,N,L}/2^L)$  for  $X \leq N < 2X$ . Let  $p_N$  be the largest prime factor of  $N + t$  and  $m_N = (N + t)/p_N$ . Define

$$\begin{aligned} \mathcal{A} &= \{N \in [X, 2X) : m_N \leq \lfloor \sqrt{X}/2 \rfloor, \quad p_N > 2\lfloor \sqrt{X} \rfloor\}, \\ \mathcal{G} &= \{N \in \mathcal{A} : \varphi(m_N) \geq \eta m_N\}. \end{aligned}$$

All index sets here contain integers. On  $\mathcal{A}$ , the inequalities ensure  $p_N > m_N \geq 1$ , so  $\varphi(N + t) = \varphi(m_N)(p_N - 1)$ . The coefficient of this totient value in  $D_{h,N,L}/2^L$  is  $(2^h - 1)/2^t$ . Set

$$z_N = \exp\left(\frac{2\pi i (2^h - 1) \varphi(m_N)(p_N - 1)}{2^t}\right), \quad w_N = E_N / z_N \quad (N \in \mathcal{A}),$$

and let  $\bar{z}_m$  be the average of  $z_N$  over  $\{N \in \mathcal{A} : m_N = m\}$ , with value zero for an empty set. The proposed bounds are

$$\begin{aligned} \operatorname{Re} \sum_{N \in \mathcal{G}} w_N (z_N - \bar{z}_{m_N}) &\leq \frac{14}{25}X, \\ \left\| \sum_{N \in \mathcal{G}} w_N \bar{z}_{m_N} \right\| &\leq \frac{1}{100}X, \\ \left\| \sum_{N \in \mathcal{A} \setminus \mathcal{G}} E_N \right\| &\leq \frac{1}{100}X, \\ \left\| \sum_{\substack{X \leq N < 2X \\ N \notin \mathcal{A}}} E_N \right\| &\leq \frac{8}{25}X. \end{aligned}$$

Before taking real parts or norms, these four sums add exactly to  $\sum_{N=X}^{2X-1} E_N$ : partition the indices and use  $w_N z_N = E_N$ . Subtracting the group mean is an identity, not an independence assumption. The four constants add to  $9/10$ , giving the preceding real-part

bound; they do not give a  $21/25$  bound on the norm of the whole sum. The allocations are sufficient margins, not claimed optimal constants. In particular,  $1 - \log 2 < 8/25$  leaves room for the smooth-number estimate used below.

**Problem 5.1** (Four estimates for the first-harmonic sum). For each  $h \geq 1$ , do there exist  $s \geq 1$  and  $0 < \eta < 1$  such that, for every  $X_0$ , there are  $X, L$  with

$$\max(X_0, 1) \leq X, \quad h \leq L - s, \quad 16(2X + h + L + 2) \leq 2^L,$$

for which those four bounds hold?

The companion gives the same decomposition, and the [four simultaneous inequalities](#) are defined in the linked source. All four bounds must hold on the *same* block with the same parameters. The permitted blocks need only be arbitrarily large, but  $L$  must also make the truncation error small enough for the residue test. The companion proves the unassigned and excluded-cofactor bounds at minimal admissible depth, using smooth-number asymptotics and the distribution of  $\varphi(m)/m$ . The unassigned bound is also a [Lean theorem](#), whose proof replaces the smooth-number asymptotics by Mertens' theorems. The excluded-cofactor bound is [proved in Lean](#) with the prime number theorem and the existence of the density of  $\{m : \varphi(m) < \eta m\}$  taken as hypotheses. An elementary mean-value estimate also gives the explicit sufficient cutoff  $\eta = 1/1000$  for the excluded-cofactor bound. The two remaining estimates concern the group means and their weighted, mean-subtracted phases. They must hold at the same depth and for the same choice of  $\eta$ ; the existence of a cutoff that controls the excluded indices does not establish either of them.

Balasubramanian, Giri and Srivastav estimate the finite sums  $\sum_{1 \leq n \leq x} F(n)G(n-h)$  for  $F = f * 1, G = g * 1$  with decaying  $f, g$ , setting  $F(n) = G(n) = 0$  for  $n \leq 0$ . Taking  $f(n) = g(n) = \mu(n)/n$  gives  $F(n) = G(n) = \varphi(n)/n$  for  $n \geq 1$ ; the estimate is uniform for  $|h| \leq x/2$  [9, Thm. 2.2]. Their partial-summation extension allows differentiable weights with bounded derivative on the summation interval [9, Rem. 2.9]. The missing input here is a useful error bound for the present phase and residual weight, not shift uniformity or partial summation itself. **Acknowledgement.** I thank Wouter van Doorn for advice on writing for a first-time reader and on explaining the strength of a hypothesis. His advice concerned a different note; it was not a mathematical review or endorsement of the results presented here.

## 6 A complete equality test at base six

The composite base shows what the integral coordinates let a reader do. At  $k = 6$  and  $e = 2$  there are 43 indexed sections. Retain  $F_{0,0}, F_{1,0}$ , the five  $F_{1,r}$  with  $1 \leq r \leq 5$ , and the 30 sections  $F_{2,r}$  with  $1 \leq r < 36$  and  $6 \nmid r$ . In addition to  $F_{2,6} = 2F_{1,1}$  and  $F_{2,12} = 4F_{1,2}$  from above, the six omitted sections reduce as

$$\begin{aligned} F_{2,0} &= 6F_{1,0}, & F_{2,6} &= 2F_{1,1}, & F_{2,12} &= 4F_{1,2}, \\ F_{2,18} &= 3F_{1,3}, & F_{2,24} &= 4F_{1,4}, & F_{2,30} &= 2F_{1,5}. \end{aligned}$$

Order the retained columns as just stated, with residues increasing within each level, and evaluate them at  $n = 0, \dots, 36$ . Exact integer totient values give a  $37 \times 37$  matrix  $M$  with  $\det M \equiv 17 \pmod{101}$ . Thus  $M$  is nonsingular over  $\mathbb{Q}$ . After the six reductions, a proposed integer identity holds for every  $n \geq 0$  precisely when its 37 retained coefficients vanish; otherwise one of those 37 inputs is an exact counterexample. The six omitted coefficients are also its unique coordinates in the relation basis of Corollary 1.3 when it is an identity. The public scripts/totient\_kernel\_normal\_form.py prints those coordinates and the first counterexample. Its modular determinant is an independence certificate; the identity decision and witness use exact integer arithmetic. For example,  $101F_{2,2}$  vanishes modulo 101 at every input, but its value at  $n = 0$  is 101, so it is not an identity.

## A Further context and formal sources

### Affine independence in the earlier literature

The separation method has an earlier source: Yazdani's proof of Theorem 2, explicitly credited there to Shallit, uses the Chinese remainder theorem and Dirichlet's theorem to distinguish kernel sections modulo a fixed modulus [10, Thm. 2 and proof, pp. 652–653]. For a later account see Allouche, Shallit and Yassawi [11, Thm. 3 and Rem. 7]. Here one separates a chosen column from every other column of a finite evaluation matrix. Nonsingularity, not merely distinctness of infinitely many sections, gives linear independence.

Martin's Theorem 1 gives another proof of the affine independence conclusion [4, Thm. 1]. For any ordering of the nonproportional forms and any  $C > 0$ , their successive totient ratios exceed  $C$  on a set of positive lower density. In a proposed relation  $\sum_i c_i \varphi(L_i(n)) = 0$ , put a nonzero coefficient first and choose  $C > \max(1, \sum_{i>1} |c_i|/|c_1|)$ . At the resulting values of  $n$ , the first term has larger absolute value than the sum of all the others, a contradiction. The proof in Section 1.1 instead constructs a finite nonsingularity certificate for each chosen family.

### Regularity, Mahler equations and section maps

The totient is not  $k$ -regular, in the sense of Allouche and Shallit [1], for any  $k \geq 2$ : Coons proved that the  $\mathbb{Z}$ -module generated by its  $k$ -kernel is not finitely generated, by counting the poles of  $\zeta(s-1)/\zeta(s)$  in the strip  $0 \leq \operatorname{Re} s \leq 1$  [2, Cor. 3.1, Thm. 3.2]. For integer-valued sequences this is equivalent to infinite dimension of the rational span: if the span has dimension  $d$ , some  $d$  evaluations map it injectively to  $\mathbb{Q}^d$  and the module into  $\mathbb{Z}^d$  (compare the evaluation argument in [3, App. A]). Theorem 1.1 therefore recovers Coons's nonregularity theorem [2, Thm. 3.2] and gives the dimension at every finite level.

The generating-function conclusion concerns functional equations, not just the dimensions of section spans. A  $k$ -Mahler equation for a formal power series  $F$  has the form

$$\sum_{i=0}^d P_i(z)F(z^{k^i}) = 0, \quad d \geq 0, \quad P_i(z) \in \mathbb{Q}[z], \quad P_0 \neq 0.$$

The height criterion of Adamczewski, Bell and Smertnig already implies that an integer-coefficient  $k$ -Mahler series with polynomially bounded coefficients is  $k$ -regular [14, Thm. 1.2(b), p. 2528]. Indeed, the logarithmic height of an integer  $a$  is  $\log \max(1, |a|)$ . Since  $0 \leq \varphi(n) \leq n$ , nonregularity rules out such an equation for  $F_\varphi(z) = \sum_{n \geq 1} \varphi(n)z^n$  in every base  $k \geq 2$ ; no multiplicativity hypothesis is needed for this deduction. The multiplicative classification in Bell and Smertnig's later preprint also gives this conclusion [12, Thm. 1.3]. Bell, Bruin and Coons give related classifications for algebraic or  $D$ -finite generating functions [13, Thms. 1.5–1.6];  $D$ -finite means satisfying a linear differential equation with polynomial coefficients. These theorems constrain the whole generating function. They do not by themselves determine the arithmetic nature of its single value  $F_\varphi(1/2)$ .

For example, the next dyadic level has rank 9 rather than 15:  $\varphi(8n+2) = \varphi(4n+1)$ ,  $\varphi(8n+4) = 2\varphi(2n+1)$  and  $\varphi(8n+6) = \varphi(4n+3)$  are its positive-residue reductions.

For a  $k$ -regular sequence that is not eventually zero, its growth exponent is the base- $k$  logarithm of the joint spectral radius of the section matrices in a basis of the kernel span. A redundant spanning family gives only an upper bound [3, Thm. 1, Prop. 4, Cor. 7]. This finite-dimensional theorem does not apply to  $\varphi$ . Its section maps nevertheless have the explicit form below. For  $0 \leq i < k$ , put  $T_i g(n) = g(kn+i)$ . These are the coefficient versions of the Cartier operators, and

$$T_i F_{j,r} = F_{j+1, k^j i + r}, \quad V_{k,e+1} = V_{k,e} + \sum_{i=0}^{k-1} T_i V_{k,e}.$$

Write each residue at level  $j+1$  as  $k^j i + r$ ,  $0 \leq r < k^j$ , to obtain the second equality. The strict dimension increase therefore prevents all  $T_i$  from preserving  $V_{k,e}$ . A retained positive-residue section maps to another such section, since  $k \nmid (k^j i + r)$  when  $j \geq 1$  and  $k \nmid r$ . The two zero-residue generators need separate treatment:  $T_i F_{0,0} = F_{1,i}$  and  $T_i F_{1,0} = F_{2,ki}$ , the latter a positive integer multiple of  $F_{1,i}$ . Thus every basis element maps to a positive integer multiple of a basis element at the next level, not necessarily to a basis element itself.

Erdős and Graham list the irrationality question for  $S$  [8, p. 61].

### A finite denominator exclusion

The companion also gives the exact Farey-interval certificate at window  $K = 240$ , excluding  $S = a/q$  for integers  $a$  and  $0 < q \leq Q_0 \approx 7.96 \times 10^{34}$ . This finite exclusion is separate from the quantified irrationality conditions.

### Source guide

Each link below opens a Lean declaration. A conditional theorem verifies an implication; its hypothesis remains unproved.

### Bases and relations.

[reduction of sections with residue zero](#); [reduction of even residues](#); [independence](#); [span equality](#); [basis](#); [independence](#); [dimension](#); [the exact finite-truncation rank](#); [infinite-dimensionality directly](#); [coordinate specification](#); [injective residues](#); [unique representation](#); [level count](#); [section-index count](#); [index as two plus the odd-residue count](#); [full](#)

index count; a nonzero evaluation determinant; an invertible minor for affine totient sections; power-multiplication identity; affine cross-gcd identity; nonzero reduction scalar; all-base residue step; canonical spanning; span equality; rank in every base from independence; all-base truncation rank from independence; independence in every integer base; the rank of the retained all-base family; the rank of all sections through a finite level.

#### **Tail identities, residue tests and carries.**

the finite-prefix and tail identity; definition; certificates for shifts 1 through 8; nonintegral tail differences at the checked shifts; the integer carry recurrence and its growth bound; residue recurrence; comparison of the two residue conditions; exact equivalence; a certificate in each sufficiently late pair of multipliers; pointwise equivalence; global equivalence; equivalence after setting depth equal to the shift; irrationality and arbitrarily large certificates; the carry-rank lower bound under rationality; carry rank at least  $2^e - 1$  from independent totient sections; modular periodicity and the rank lower bound for the same carry; forcing vanishing; geometric reduction; identity; sign; primes; prime powers; unboundedness; exact centre decomposition; explicit error bound; separation criterion; target equivalence; endpoint nonnegativity; foreign-channel identity; five-point loss bound; simultaneous finite family; finite horizon; vanishing block; nonzero block; centrality equivalence; equivalence of the two conditions; conditional irrationality theorem; finite complement bound; exact channel split; conditional target-miss theorem; the periodic theorem;  $\sum \varphi(n)/(2^n - 1) = 2$ ; rationality and integer carries with controlled growth; diagonal split; definition; definition of the scaled tail; the proposed cyclotomic residue condition; the checked equivalence; checked conditional support/period counterexample; the checked binary-layer theorem; certificates with depth equal to the period multiple.

#### **Finite denominator exclusions.**

exclusion; reduced-denominator form; first failure; the conditional implication; farey gap; the denominator bound at window 240; denominator exclusion from a finite interval; the earlier denominator bound at window 120; denominator exclusion for the coprimality probability; list; verification; certificates at every scale through 82.

#### **Least common multiples and limits of the tested methods.**

certificate completeness; tail-difference integrality; irrationality iff every tail difference is nonintegral; exact frontier; forward implication; positive tail differences; negative endpoint representative; residue near the upper endpoint; divisibility preserved by finite combinations of shifts; the counterexample with sparse prescribed values; pointwise; cofinal at each shift; lcm diagonal; the certificate at  $t = 2^4$  and depth 23; the certificate at  $t = 2^6$  and depth 93; actual-orbit non-integrality at both heights; certificates beyond every threshold  $a_0 \leq 6$ ; an explicit approximation error; the approximation as an integer over  $4^q$ ; distance at least  $1/32$  from every integer; the cofinal separation hypothesis; the conditional endpoint theorem; the staircase is impossible under the stated room hypothesis; terminal-remainder identity; the limitation of fixed-precision valuation data; rationality from an integral tail difference.

#### **Möbius identities and rational approximation.**

unique minimizer; uniform lower bound  $21/320$ ; lower bound  $1/16$ ;  $1/15$  failure; positive weighted averages of the quotients; rational linear-form obstruction; divisor-count layer; first gcd moment; no fixed index; the primorial divisibility; irrationality of the Erdős–Borwein series;  $\sum \mu(n)/(2^n - 1) = 1/2$ ; the squared-Mersenne expression for the totient series; the coprimality-probability expression; the Lambert-weighted sum over coprime positive pairs; the Stern–Brocot recursion with stopping;  $s = 1$ ;  $s = 2$ ; first difference; second difference.

### Proposed exponential-sum and prime-factor estimates.

window-separated pairs; the unproved quantitative separation condition at primes; a positive adaptive truncation budget;  $S$  is irrational; density hypothesis; block-gap hypothesis; density-to-gap implication; exact squaring recurrence; iterate formula; initial-phase normal form; initial-phase equivalence; dyadic-root obstruction; prime-alignment implication; mixed difference; separable case; fixed-stencil uniqueness; a four-point divisor-layer identity; a centred representative for a residue class modulo an even modulus; eventual nontriviality and coprimality of the layers; order divisibility with bounded extension degree; exclusion of any fixed finite prime support; unbounded prime divisors; membership; injectivity; image equality; checked counterexample; the exact decomposition into four prime-factor contributions; the conditional irrationality theorem; the four inequalities on a common block; the first-harmonic bound from those four inequalities.

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